Module	**SE101.3 -OOP with Java – 25.3**
Lab sheet Number	**08**
Areas covered	**Revision**

Question No 01

Create a class called **Person** with suitable access modifiers and data types to store person ID, name, and age. Write a parameterized constructor and getters/setters for all variables.

a. Create a subclass called **Student** that has an Is-a relationship with Person. Add a variable called course and write a parameterized constructor.

b. Write a method called **displayDetails()** in Person class that prints all person details.

c. Perform **displayDetails()** in Student class and print person details along with course.

d. In main(), create a Person object and a Student object. Call displayDetails() on both objects and print the results.

Question 02

Create a class called Vehicle with suitable access modifiers and data types to store vehicle ID, brand, and speed. Write a parameterized constructor and getters/setters for all variables.

a. Create a subclass called Car that has an Is-a relationship with Vehicle. Add a variable called numberOfDoors and write a parameterized constructor.

b. Write a method called calculateSpeed() in Vehicle class.

c. Perform calculateSpeed() in Car class and print a suitable message.

d. In main(), create a Vehicle reference variable and assign a Car object to it. Call calculateSpeed() and print the result.

Question 03

Create a class called Employee with suitable access modifiers and data types to store employee ID, employee name, and salary. Write a parameterized constructor and getters/setters for all variables.

a. Create subclasses called Manager and Developer that have an Is-a relationship with Employee. Add department for Manager and programmingLanguage for Developer.

b. Write a method called calculateBonus() in Employee class.

c. Perform calculateBonus() in both subclasses using suitable bonus calculations.

d. In main(), create Manager and Developer objects. Call calculateBonus() and print the results.

Question 04

Create a class called Person with suitable access modifiers and data types to store person ID and name. Write a parameterized constructor and getters/setters for all variables.

a. Create a subclass called Employee that has an Is-a relationship with Person. Add salary and write a parameterized constructor.

b. Create another subclass called Manager that has an Is-a relationship with Employee. Add department and write a parameterized constructor.

c. Write a method called displayInfo() that displays all information.

d. In main(), create a Manager object and display all details.

Question 05

Create a class called Payment with suitable access modifiers and data types to store payment ID, customer name, and amount. Write a parameterized constructor and getters/setters for all variables.

a. Create subclasses called CashPayment, CardPayment, and OnlinePayment. Add suitable variables for each subclass.

b. Write a method called processPayment() in Payment class.

c. Perform processPayment() in all subclasses and print suitable messages.

d. In main(), create a Payment reference variable and assign different payment objects to it. Call processPayment() and print the results.